Unit 2

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# McCulloch and Pits Neural Network.

(MCP Model) → M-P Neuron

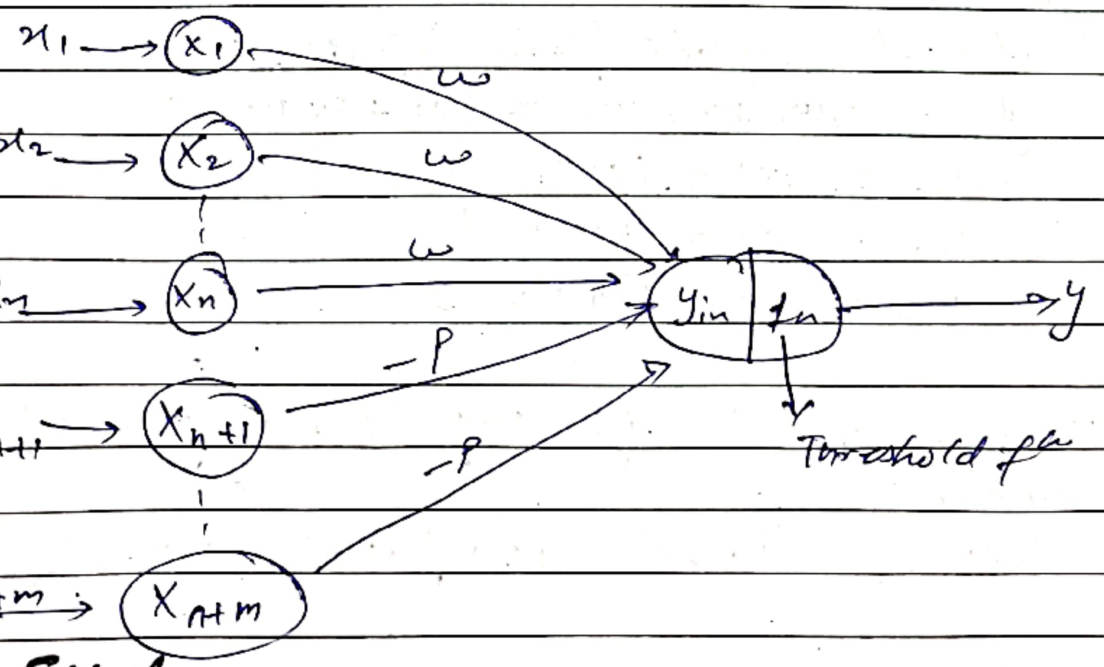
↳ one of the earliest mathematical model of ANN.
↳ Proposed by "Warren McCulloch and Walter Pitts"
↳ in 1943
↳ works on Threshold activation fn.
↳ MCP model is a "Binary threshold neuron"
i.e, takes Binary Input → mostly used in logical fn.
↳ can Implement AND, OR and NOT Gate,
↳ only Two case - Either neuron may fire or may not fire.
↳ weight may be → Excitatory (+ve)
→ Inhibitory (-ve)
↳ Threshold plays an important role in MCP model.

$$y = \begin{cases} 1 & , x \geq \theta \\ 0 & , x < \theta \end{cases}$$

$\theta$ = Threshold value

**Architecture**

↳ Input → Binary Input.
$x_i \in \{0, 1\}$




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$\theta \geq nw - p$

w = excitatory weight

p = inhibitory weight

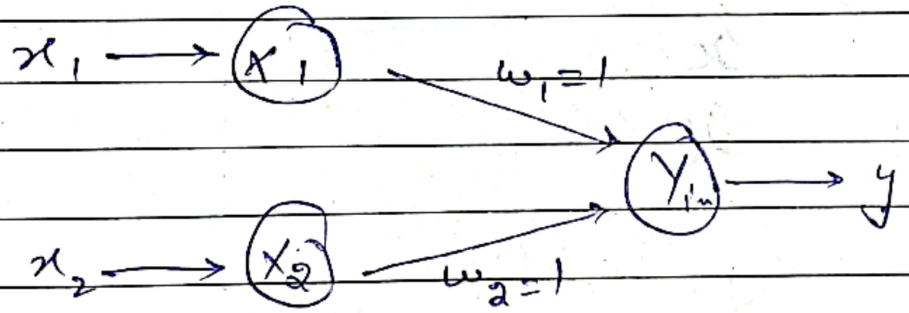
## # MCP Model for AND function

Truth Table of AND f^n ⇒

| $x_1$ | $x_2$ | y |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 0 |

$Y = X_1$ AND $X_2$

Assuming the weights be $w_1 = 1$ and $w_2 = 1$




With the assumed weight, net input is calculated for the four inputs —

$(\overset{x_1}{1}, \overset{x_2}{1})$, $y_{in} = x_1 w_1 + x_2 w_2 = 1 \times 1 + 1 \times 1 = 2$

$(1,0)$, $y_{in} = 1 \times 1 + 1 \times 0 = 1$

$(0,1)$, $y_{in} = 1 \times 0 + 1 \times 1 = 1$

$(0,0)$, $y_{in} = 0 \times 1 + 0 \times 1 = 0$

For AND f^n, output is high only if both the inputs are high,

For this condition net input is calculated as

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Hence, based on the net input $\theta$ is set to 2.

$\theta \geq nw - p$ $\quad$ [$n = 2, w = 1$ (excitatory weights) $\quad p = 0$ (no inhibitatory weights)]

$\geq 2 \times 1 - 0 \geq 2.$

$$y = \begin{cases} 1 & \text{if } y_{in} \geq 2. \\ 0 & \text{if } y_{in} < 2. \end{cases}$$

## # MCP Model for OR function

Truth Table for OR fn ⇒

| $x_1$ | $x_2$ | $y$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

Let, $w_1 = 1$, $w_2 = 1$. $\quad \theta = 1$

$x_1$ $x_2$

(0, 0), $y_{in} = 0 \times 1 + 0 \times 1 = 0$

(0, 1), $y_{in} = 0 \times 1 + 1 \times 1 = 1$

(1, 0), $y_{in} = 1 \times 1 + 0 \times 1 = 1$

(1, 1), $y_{in} = 1 \times 1 + 1 \times 1 = 2$

$$y = \begin{cases} 1 & \text{if } y_{in} \geq 1 \\ 0 & \text{if } y_{in} < 1 \end{cases}$$

# MCP model for NOT function

| x | y |
|---|---|
| 0 | 1 |
| 1 | 0 |

Let Assum $w = -1$ $\quad \theta = 0$

(0) , $y_{in} = 0 \times (-1) = 0$

(1) , $y_{in} = 1 \times (-1) = -1$

$$y = \begin{cases} 1 , & x \geq 0 \\ 0 , & x < 0 \end{cases}$$

| | MCP | Hebb |
|---|---|---|
| Primary focus. | Neuron Activation | Weight adjustment / learning |
| Weights | Predetermined/fixed | Modified during training |
| Learning | No Learning mechanism | Hebb Learning Rule. |
| Training. | Not Required | Required. |
| main focus | Models the operation of a neuron | models the learning process of a neural network. |

→ uses Hebb Learning Rule.

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# Hebb Model

MCP - 1943 (1)

→ by Donald O. Hebb in 1949 (2)

→ "Neuron that fire together, wire together"

**Basic idea :-** When two neurons are active at the same time, the connection btw them becomes stronger.

→ The Hebb Model therefore learns by changing the **weights** btw neurons.

→ represent the strength of the connection btw input and an output.

$w_i(new) = w_i(old) + x_i d$

→ Hebb rule is suitable for bipolar data (-1, 1) than binary data

→ If binary data is used, then above weight updation formula can't distinguish two conditions :-

(1) Training pair in which input unit is "on" and target value is "off"

(2) Training Pair in which both the input unit and target value are "off"

**Fundamental Idea :-** $\Delta w_i = x_i d$

where,

$x_i$ = input

$d$ = desired / output response.

$\Delta w_i$ = change in weight.

$\therefore w_i(new) = w_i(old) + x_i d$

**For bias :-** $\Delta b = d$

$\therefore b(new) = b(old) + d$

**Means -**

① $x_i$ and $d$ → same sign
↳ weight ↑

② $x_i$ and $d$ has opposite sign
↓
weight ↓

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# Training of Hebb Network.

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↳ Training means adjusting the weights according to the training example.

**Step 1 :-** Initialize weights and bias — to zero

$$w_1 = w_2 \text{ ---} = w_n = 0.$$

$$b = 0$$

**Step 2 :-** Provide input Pattern and Desired output

$X = (x_1, x_2 \text{ ---} x_n)$ and $d$.

**Step 3 :-** calculate weight change.

$\Delta w_i = x_i d$ and $\Delta b = d$

**Step 4 :-** Update Weights.

$w_i(\text{new}) = w_i(\text{old}) + \Delta w_i$

$(\text{new})b = b(\text{old}) + \Delta b.$

**Step 5 :-** Repeat the process for every training pattern. After training, the final weights are used during testing.

**Testing of Hebb Network.** — Testing means checking whether the trained network produces the correct output for an input.

(1) For a test input — $X(x_1, x_2, x_3 - - x_n)$

(2) calculate — $net = \sum_{i=1}^{n} x_i w_i + b.$

(3) Apply activation $f^n$.

for Bipolar output :-

$$y = \begin{cases} +1, & net > 0 \\ 0, & net = 0 \\ -1, & net < 0 \end{cases}$$

**Spiral** The obtained output is compared with the expected output

# Hebb Network for AND fn

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AND Gate :-

| $x_1$ | $x_2$ | output |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

| $x_1$ | $x_2$ | d |
|---|---|---|
| -1 | -1 | -1 |
| -1 | +1 | -1 |
| +1 | -1 | -1 |
| +1 | +1 | +1 |

Binary to Bi-polar conversion :- $0 \longrightarrow -1$

$1 \longrightarrow +1$

Initialize → $w_1 = 0$, $w_2 = 0$, $b = 0$

(1) Pattern 1 :- $x_1 = -1$, $x_2 = -1$, $d = -1$

$\therefore \Delta w_1 = (-1)(-1) = 1.$

$\Delta w_2 = (-1)(-1) = 1.$

$\Delta b = -1$

update Weight :- $w_1 = 0 + 1 = 1$

$w_2 = 0 + 1 = 1$

$b = 0 - 1 = -1$

(2) Pattern 2 :- $x_1 = -1$, $x_2 = +1$, $d = -1$

$\Delta w_1 = (-1)(-1) = 1$

$\Delta w_2 = (1)(-1) = -1$

$\Delta b = -1$

updated weight :- $w_1 = 1 + 1 = 2$

$w_2 = -1 + 1 = 0$

$b = -1 - 1 = -2.$

(3) Pattern 3 :- $x_1 = +1$, $x_2 = -1$, $d = -1$

$\Delta w_1 = 1 \times -1 = -1$

$\Delta w_2 = -1 \times -1 = 1.$

$\Delta b = -1$

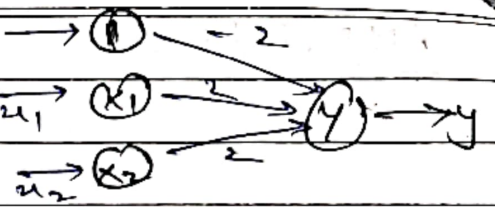
Updated Weight :- $w_1 = 2 - 1 = 1$

$w_2 = 0 + 1 = 1$

$b = -2 - 1 = -3$

$w_1 = 1, \quad w_2 = 1, \quad b = -3$

(4) Pattern :- $x_1 = +1, \quad x_2 = +1, \quad d = +1$




$\Delta w_1 = 1 \times 1 = 1$

$\Delta w_2 = 1 \times 1 = 1$

$\Delta b = 1$

Updated weight = $w_1 = 1+1 = 2$

$w_2 = 1+1 = 2$

$b = -3+1 = -2$

Thus, After Training —

$w_1 = 2, \quad w_2 = 2, \quad b = -2$

**Testing**

Final eqⁿ ⇒

$net = 2x_1 + 2x_2 - 2$

Activation fⁿ ⇒

$$y = \begin{cases} +1, & net \geq 0 \\ -1, & net < 0 \end{cases}$$

Test 1 $(-1, -1)$ :

$net = 2(-1) + 2(-1) - 2 = -6$

$\therefore \quad y = -1$

Test 2 $(-1, +1)$ :

$net = 2(-1) + 2(1) - 2 = -2$

$\therefore \quad y = -1$

Test 3 $(+1, -1)$ :

$net = 2(1) + 2(-1) - 2 = -2$

$\therefore \quad y = -1$

Test 4 $(+1, +1)$ :

$net = 2(1) + 2(1) - 2 = 2$

$\therefore \quad y = +1$

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## Hebb Network for ~~AND~~ OR f^n :-

| $x_1$ | $x_2$ | d |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

| $x_1$ | $x_2$ | d |
|---|---|---|
| -1 | -1 | -1 |
| -1 | +1 | +1 |
| +1 | -1 | +1 |
| +1 | +1 | +1 |

Initialize, $w_1 = 0$, $w_2 = 0$, $b = 0$.

(1) $x_1 = -1$, $x_2 = -1$, $d = -1$.

$\Delta w_1 = (-1)(-1) = 1$.

$\Delta w_2 = (-1)(-1) = 1$.

$\Delta b = -1$

updated weights :- $w_1 = 1$, $w_2 = 1$, $b = -1$

(2) $x_1 = -1$, $x_2 = +1$, $d = +1$

$\Delta w_1 = (-1)(1) = -1$

$\Delta w_2 = (+1)(1) = 1$.

$\Delta b = 1$

updated weights :- $w_1 = 1-1 = 0$, $w_2 = 1+1 = 2$, $b = -1+1 = 0$

$w_1 = 0$, $w_2 = 2$, $b = 0$

(3) $x_1 = +1$, $x_2 = -1$, $d = +1$

$\Delta w_1 = 1 \times 1 = 1$ $\quad w_1 = 0+1 = 1$

$\Delta w_2 = -1 \times 1 = -1$ $\quad w_2 = 2-1 = 1$

$\Delta b = 1$ $\quad b = 0+1 = 1$

updated weights :- $w_1 = 1$, $w_2 = 1$, $b = 1$.

(4) $x_1 = +1$, $x_2 = +1$, $d = +1$

$\Delta w_1 = 1 \times 1 = 1$ $\quad w_1 = 1+1 = 2$.

$\Delta w_2 = 1 \times 1 = 1$ $\quad w_2 = 1+1 = 2$

$\Delta b = 1$ $\quad b = 1+1 = 2$.

∴ Final :- $w_1 = 2$, $w_2 = 2$, $b = 2$

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Test Phase :1

eqn ⇒ $y = 2x_1 + 2x_2 + 2$

(1) $(-1,-1)$ : $-2-2+2 = -2$.

$\therefore \boxed{y = -1}$

(2) $(-1,+1)$ : $-2+2+2 = 2$

$\therefore \boxed{y = +1}$

(3) $(+1,-1)$ : $2-2+2 = 2$.

$\therefore \boxed{y = +1}$

(4) $(+1,+1)$ : $2+2+2 = 6$

$\therefore \boxed{y = +1}$

Activation fn :-

$$y = \begin{cases} +1 , & y > 0 \\ 0 , & y = 0 \\ -1 , & y < 0 \end{cases}$$

→ Perceptron learning Rule

# Perceptron Network

MCP → 1943
Hebb → 1949

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→ by Frank Rosenblatt in 1957
→ Works on error

→ Perceptron Network consist of 3 units :-
1. Sensory unit (input unit)
2. Associate unit (hidden unit)
3. Response unit (output unit)

NOTE: ① Binary Activation $f^n$ is used in sensory and Associate unit. (↳ 0, 1)

② Binary step $f^n$ (−1, 0, 1) is used in Response unit

$$\therefore \text{output} = \begin{cases} 1, & net > 0 \\ 0, & net = 0 \\ -1, & net < 0 \end{cases}$$

$y \neq d$

(e) error = d (desired output) − y (Actual output)

$$\therefore \boxed{e = d - y}$$

$$\therefore \boxed{w_i(new) = w_i(old) + \eta e x_i}$$

↳ learning Rate (Generally 1)

$$b(new) = b(old) + \eta e$$

→ If no errors, no weight updation.

# Perceptron Network for AND f^n

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| $x_1$ | $x_2$ | d |
|---|---|---|
| -1 | -1 | -1 |
| -1 | +1 | -1 |
| +1 | -1 | -1 |
| +1 | +1 | +1 |

Initialize, $w_1 = 0$, $w_2 = 0$, $b = 0$

(1) $x_1 = -1$, $x_2 = -1$, $b = 0$, $w_1 = 0$, $w_2 = 0$

net = $(-1)(0) + (-1)(0) + 0 = 0$

$y = 0$, $d = -1$ $\boxed{y \neq t}$

$\therefore e = d - y = -1 - 0 = -1$

$\therefore \Delta w_1 = (-1)(-1) = 1$

$\Delta w_2 = (-1)(-1) = 1$. $\boxed{w_1 = 1, w_2 = 1, b = -1}$

$\Delta b = -1$

(2) $x_1 = -1$, $x_2 = +1$

net = $(-1)(1) + 1 \times 1 - 1 = -1 + 1 - 1 = -1$.

$y = 0$, $d = -1$

$y \neq d$ ( need to update weight )

$\therefore e = -1 - 0 = \boxed{e = -1}$

$\Delta w_1 = 1$., $\Delta w_2 = -1$, $\Delta b = -1$

Update weight :-

$\therefore w_1 = 2$, $w_2 = 0$, $b = -2$

(3) $x_1 = +1$, $x_2 = -1$

net = $2 + 0 - 2 = 0 \Rightarrow y = 0$, $d = -1$

$y \neq d$.

$\therefore e = -1 - 0 = -1$

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$\therefore \Delta w_1 = -1$
$\Delta w_2 = 1$
$\Delta b = -1$

update weight
$w_1 = 1, w_2 = 1$
$b = -3$

(4) $x_1 = 1, x_2 = 1$
net = $1 + 1 - 3 = -1$
$\therefore y = 0, d = +1 \quad y \neq d$
$\therefore e = 1 - 0 = \boxed{1 = e}$
$\therefore \Delta w_1 = 1$
$\Delta w_2 = 1$
$\Delta b = 1$
update weights :—
$w_1 = 2, w_2 = 2, b = -2.$

exm $y = 2x_1 + 2x_2 - 2.$

Testing
$(-1, -1) \rightarrow -2 - 2 - 2 \rightarrow -6 \quad \therefore \boxed{y = -1}$
$(-1, +1) \rightarrow -2 + 2 - 2 \rightarrow -2 \quad \therefore \boxed{y = -1}$
$(+1, -1) \rightarrow +2 - 2 - 2 \rightarrow -2 \quad \therefore \boxed{y = -1}$
$(+1, +1) \rightarrow +2 + 2 - 2 \rightarrow +2 \quad \therefore \boxed{y = +1}$

**Perceptron Network for OR fn**

Initialize ; $w_1 = 0, w_2 = 0, b = 0$

(1) $x_1 = -1, x_2 = -1$
net = $-1 \times 0 + 0 \times (-1) + 0 = 0$
$\therefore y = 0$ and $d = -1$ $(y \neq d)$
$\therefore e = d - y = -1 - 0 = -1$
$\therefore \Delta w_1 = e x_1 = (-1)(-1) = 1$
$\Delta w_2 = (-1)(-1) = 1$
$\Delta b = -1$

| $x_1$ | $x_2$ | d |
|---|---|---|
| -1 | -1 | -1 |
| -1 | +1 | +1 |
| +1 | -1 | +1 |
| +1 | +1 | +1 |

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updated weights :—

$w_1 = 1, w_2 = 1, b = -1$

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(2) $x_1 = -1, x_2 = +1$

$net = -1 \times 1 + 1 \times 1 - 1 = -1 \Rightarrow y = 0$ & $d = +1$

$y \neq d \rightarrow e = 1 - 0 = 1$

$\therefore \Delta w_1 = 1 \times (-1) = -1$

$\Delta w_2 = 1 \times 1 = 1$

$\Delta b = 1.$

updated weights :—

$w_1 = 0, w_2 = 2, b = 0$

(3) $x_1 = +1, x_2 = -1$

$net = 1 \times 0 - 2 \times 1 + 0 = -2 \Rightarrow y = 0$ & $d = +1$

$y \neq d \rightarrow e = 1 - 0 = 1.$

$\therefore \Delta w_1 = 1 \times 1 = 1$

$\Delta w_2 = 1 \times (-1) = -1$

$\Delta b = 1$

updated weights :—

$w_1 = 1, w_2 = 1, b = 1$

(4) $x_1 = +1, x_2 = +1$

$net = 1 \times 1 + 1 \times 1 + 1 = 3 \Rightarrow y = +1$ and $d = +1$

(No need to update weights) $y = d.$

$\therefore$ Final weights $\Rightarrow$ $w_1 = 1, w_2 = 1, b = 1$

Testing Phase :— $\boxed{y = x_1 + x_2 + 1}$

- $(-1, -1)$ : $-1 - 1 + 1 = -1$ $\therefore \boxed{y = -1}$
- $(-1, +1)$ : $-1 + 1 + 1 = +1$ $\therefore \boxed{y = +1}$
- $(+1, -1)$ : $1 - 1 + 1 = +1$ $\therefore \boxed{y = +1}$
- $(+1, +1)$ : $1 + 1 + 1 = +3$ $\therefore \boxed{y = +1}$

Activation f$^{n}$ :-

$$y = \begin{cases} +1, & y > 0 \\ 0, & y = 0 \\ -1, & y < 0 \end{cases}$$